

Lab Assignment 2 – Student Management REST API

Web Dev III (Node.js & Express Backend)

Unit-2 | Marks: 2.5 | In-Class Lab

Assignment Overview

The Student Management REST API is a practical backend assignment where you will build a REST API using Express.js. You will manage student records using CRUD operations and learn routing, middleware, REST APIs, and error handling.

Learning Objectives

- Understand Express.js server setup
- Build REST APIs
- Perform CRUD operations
- Use middleware
- Handle errors and status codes
- Test APIs using Postman

Technology Stack

- Node.js
- Express.js
- Postman

Restrictions

No Database (MongoDB/MySQL)

No Mongoose

Use Array and JSON Data Only

Functional Requirements

1. Create Express Server
2. Create Student CRUD APIs
3. Implement Custom Logger Middleware
4. Use Modular Routing
5. Handle Errors Properly
6. Test APIs using Postman

Project Structure

Create folders:

- routes
- middleware
- data

Files:

- app.js
- studentRoutes.js
- logger.js
- students.js

Required APIs

GET /students

GET /students/:id

POST /students

PUT /students/:id

DELETE /students/:id

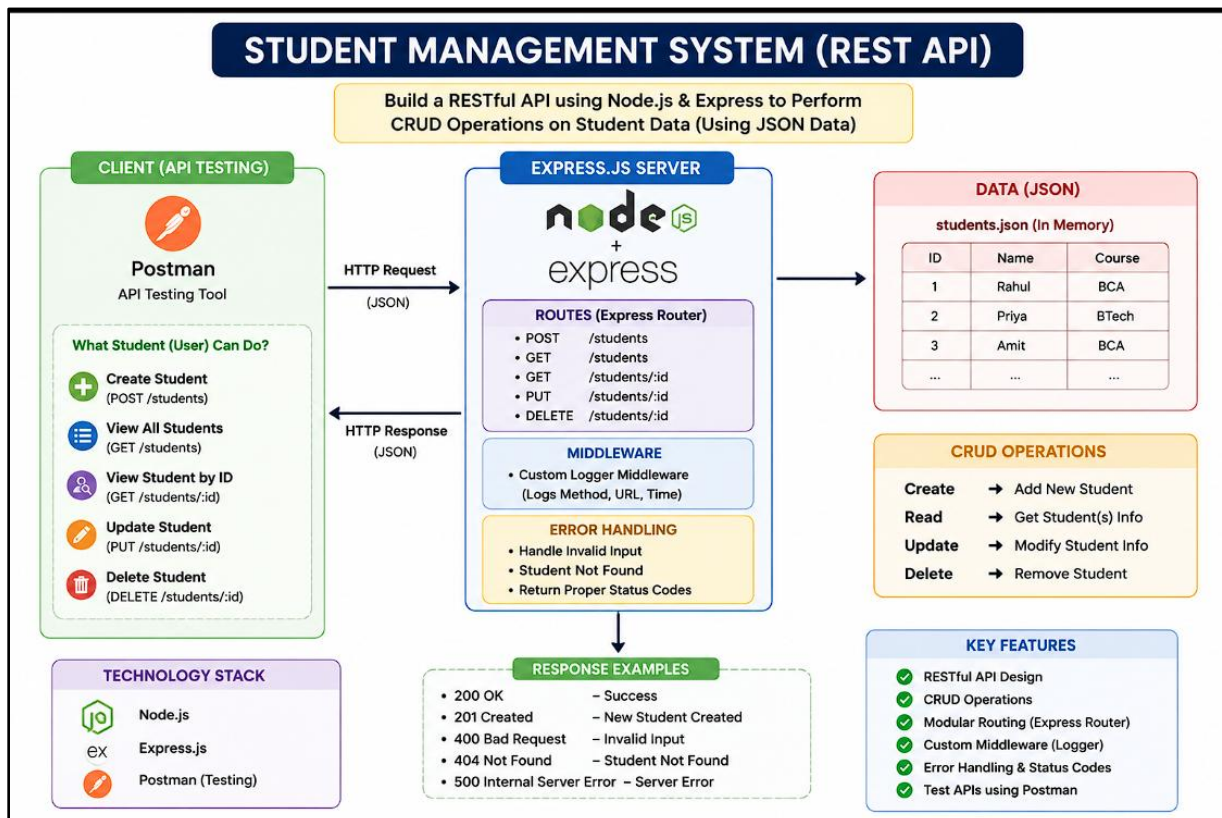
Error Handling

Return proper status codes:

- 200 Success
- 201 Created
- 400 Bad Request
- 404 Not Found

Expected Output

Students should be able to add, view, update, and delete student records using REST APIs.



Submission Guidelines

- Complete all APIs
- Use modular routing
- Use middleware
- Test all APIs in Postman
- Submit project GitHub Link

Rubric

Rubric for Assignment - 2

Criteria	Marks
Functionality	1.5
API Design	0.5
Clean Code	0.5
Total	2.5

Happy Coding! 🎯